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DUNMAN HIGH SCHOOL Preliminary Examination Year 6

H1 BIOLOGY

8876/01

Paper 1 Multiple Choice Questions

27 September 2021

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your centre number, index number, name and class at the top of this page.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

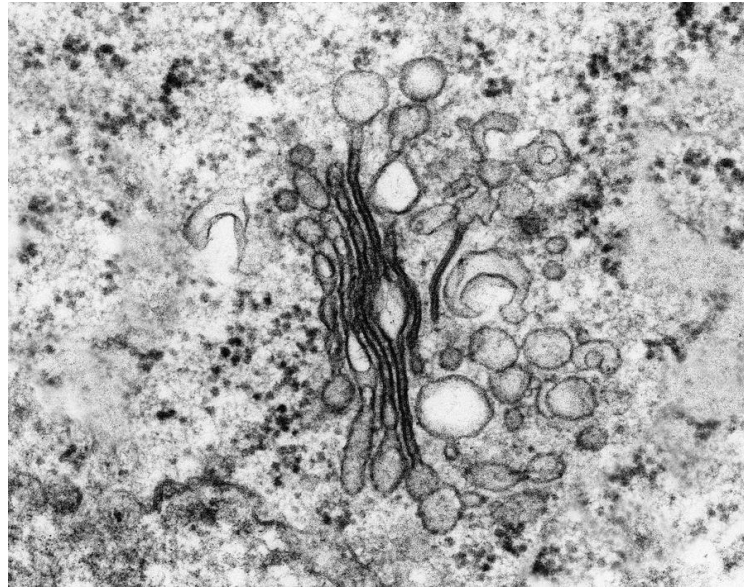
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **20** printed pages (including this cover page) and **0** blank page.

Answer **all** questions.

- 1 Figure below shows an electron microscopic image from part of an eukaryotic cell showing several organelles.



Which of the following is **not** a function of the organelles shown in the figure?

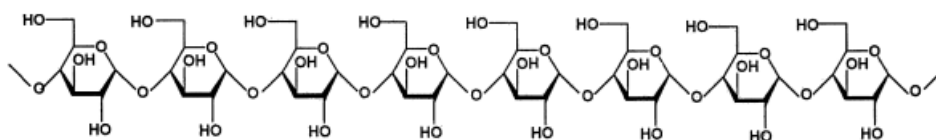
- A Store excess glucose to maintain cytosolic osmotic pressure
- B Chemically modify proteins with the addition of glycan chains
- C Carry transport proteins to the cell surface membrane
- D Digest ingested materials

2 Which statements correctly describe endoplasmic reticulum?

- 1 It contains enzymes for synthesis of polypeptides.
- 2 It has a perforated membrane.
- 3 It has a high concentration of ions like calcium.
- 4 Its membrane is constantly forming and breaking.

- A 1 and 2 only
B 1, 3 and 4 only
C 2, 3 and 4 only
D 1, 2 and 3 only

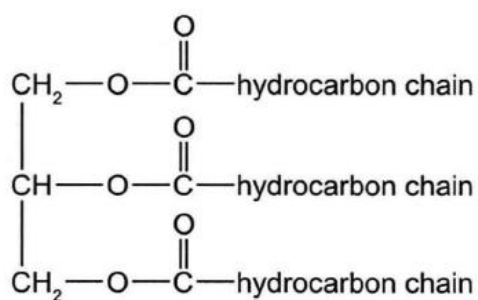
3 The figure below shows part of a biomolecule.



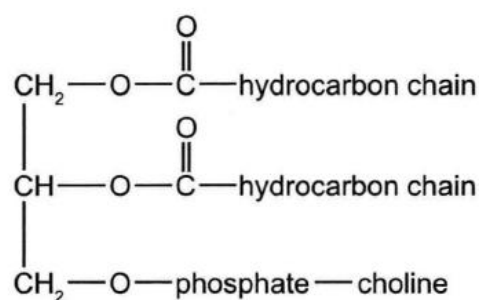
Which statement correctly describes this molecule?

- A This molecule is soluble in water.
B The strand exist as a straight chain.
C The monomers are joined together by ester bonds.
D This molecule stains iodine blue-black.

4 The diagram shows the structure of two lipid molecules.



X



Y

Which row correctly matches the identification of X and Y to their description?

	Identify X	Describe X	Identify Y	Describe Y
A	triglyceride	Made by condensation reactions with the removal of 3 water molecules.	phospholipid	Amphipathic due to presence of phosphate – choline and hydrocarbon chain.
B	triglyceride	Hydrocarbon chains do not contain C=C double bonds.	phospholipid	Hydrocarbon chains contain C=C double bonds to regulate membrane fluidity.
C	phospholipid	Made by condensation reactions with the removal of 3 water molecules.	triglyceride	Hydrocarbon chains contain C=C double bonds to regulate membrane fluidity.
D	phospholipid	Hydrocarbon chains do not contain C=C double bonds.	triglyceride	Amphipathic due to presence of phosphate – choline and hydrocarbon chain.

- 5 A student was performing food tests on some solutions and he recorded his observations in the table below. However, he forgot to include headings in his table and what each row and column represented.

	Test P	Test Q	Test R
Solution A	Brick-red precipitate	Brick-red precipitate	Remained blue
Solution B	Remained blue	Brick-red precipitate	Remained blue
Solution C	Brick-red precipitate	Brick-red precipitate	Turned violet

Which statement is most likely correct?

- A** All 3 solutions contain reducing sugars.
- B** Solution C contains two types of substrates.
- C** All 3 tests were performed with boiling water bath.
- D** Solution C contains starch and amylase.
- 6 The following are the types of bonds present in a protein.
- 1 Peptide bond
 - 2 Hydrogen bond
 - 3 Hydrophobic interaction
 - 4 Ionic bond
 - 5 Disulfide linkage

Which row correctly identified the bonds associated with the type of protein?

	Protein released immediately from ribosome	Protein globular in shape	Protein expressed in red blood cells capable of transporting oxygen
A	1	1, 2, 3, 4, 5	1, 2, 3, 4
B	2	1, 2, 3, 4, 5	1, 2, 3, 4
C	1	2, 3, 4, 5	2, 3, 4
D	2	2, 3, 4, 5	4, 5

- 7 The diagram shows a metabolic pathway.



Each stage is catalysed by one of the three enzymes: E1, E2 and E3.

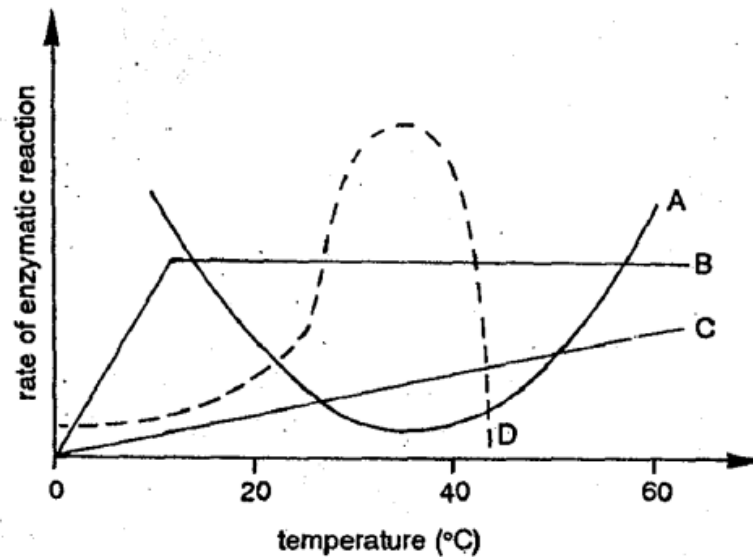
Combinations of these enzymes were added to solutions of substrate J and the products were identified.

Enzyme added to a solution of substrate J	Product
E2	K
E2 and E3	M
E1 and E2	K
E1, E2 and E3	N

Which row shows a possible sequence of enzyme activity in this metabolic pathway?

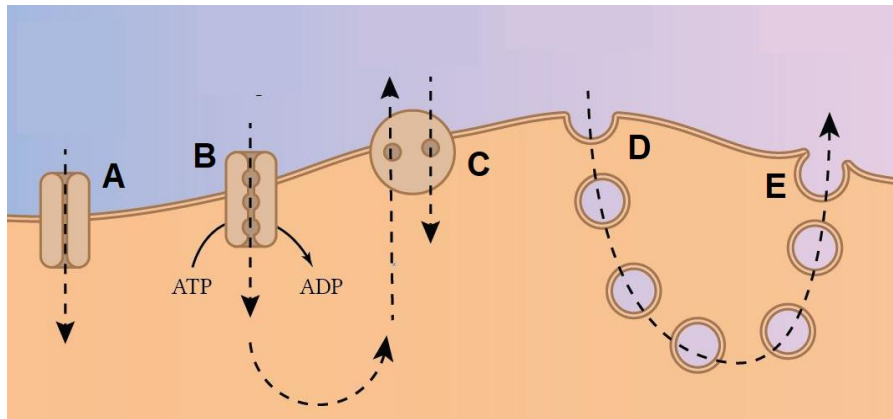
	J → K	K → L	L → M	M → N
A	E2	E2	E3	E1
B	E2	E3	E2	E1
C	E3	E1	E2	E1
D	E3	E2	E1	E3

- 8 Which of the following curves best illustrates the effect of temperature on the rate of an enzymatic reaction?



- 9 Which statement correctly relates the structure of cholesterol to its role in the membrane?
- A It has C=C double bonds to regulate the membrane fluidity.
 - B It is amphipathic allowing it to interact with the phospholipid molecules and serve as a molecular plug to regulate transport across the membrane.
 - C The OH group present in its structure enables it to function as an organic solvent for transporting molecules across the membrane.
 - D It has hydrophobic fatty acid tails that interact with the phospholipid tails, holding them close at high temperature and preventing the membrane from breaking.

- 10 The figure below shows different mechanisms of transport across membrane, labelled A to E.



Which of the following shows a correct organisation of the mechanisms according to their features?

A

Passive	Active
A, C	B, C, D, E

B

Transport polar molecules	Transport non-polar molecules
A, B, C	D, E

C

Transport molecules down concentration gradient	Transport molecules against concentration gradient
A, C	B, C, D, E

D

Transport large molecules	Transport small molecules
A, B, C	D, E

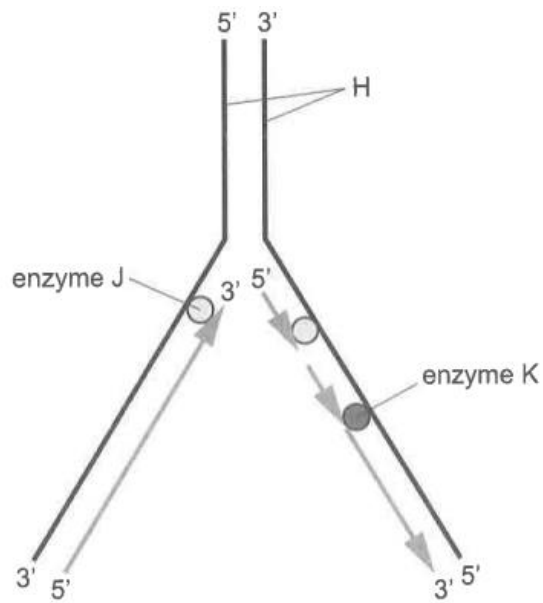
11 The table shows the percentages of bases in DNA from various sources.

source of DNA	adenine	guanine	thymine	cytosine
calf thymus	28.2	21.5	27.8	22.5
bull spleen	27.9	22.7	27.3	22.1
bull sperm	28.7	22.2	27.2	22.0
rat bone marrow	28.6	21.4	28.4	21.5
yeast	31.3	18.7	32.9	17.1

What is a valid deduction from the data?

- A** All cells from the same species have about the same amount of DNA.
- B** The four bases show complementary base pairing.
- C** The ratio of purine to pyrimidine in yeast cells is higher than animal cells.
- D** DNA structure is linear in both yeast and animal cells.

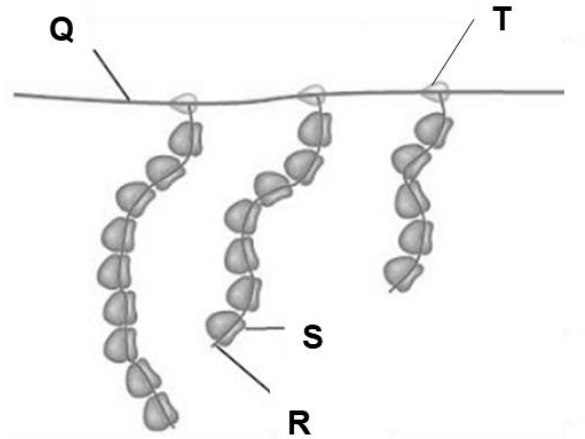
12 The figure below shows DNA replication.



Which row correctly describes the labelled features on the figure?

	Name of structures labelled H	Name of enzyme J	Function of enzyme K
A	Polynucleotide strand	DNA polymerase I	Replaces RNA primer with DNA
B	Polynucleotide strand	DNA polymerase III	Joins DNA fragments
C	Sugar phosphate backbones	DNA ligase	Joins nucleotides
D	Sugar phosphate backbones	DNA polymerase III	Synthesises DNA strand

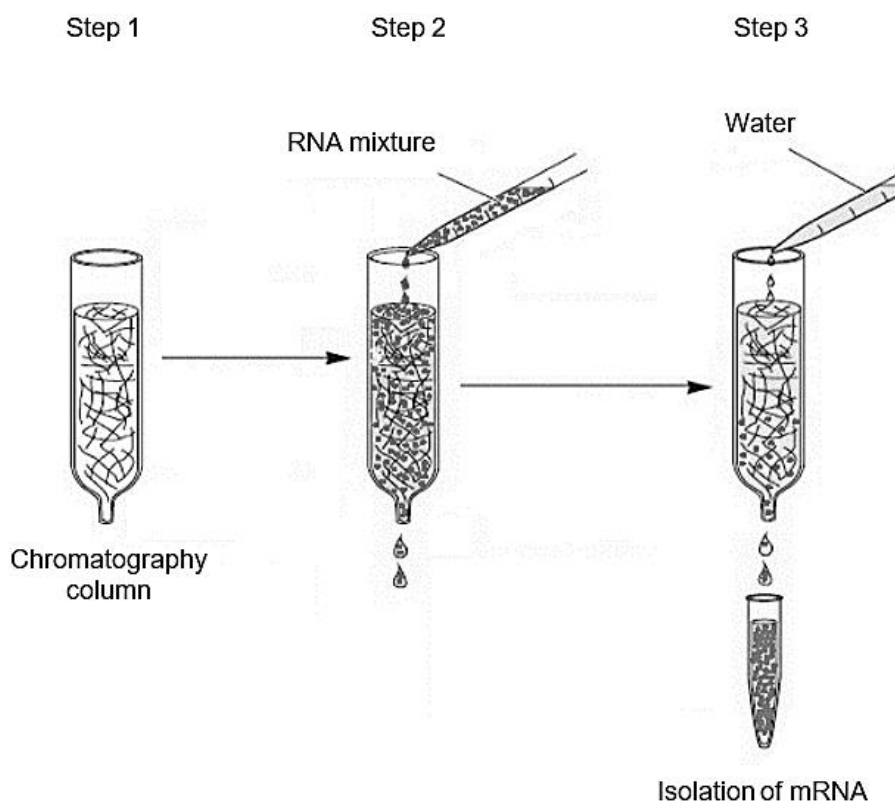
- 13** The figure below shows how transcription and translation occur simultaneously in prokaryotes.



Which row correctly names the various labelled components?

	Q	R	S	T
A	DNA	RNA	Ribosome	RNA polymerase
B	RNA	DNA	Ribosome	DNA polymerase
C	RNA	polypeptide	RNA polymerase	Ribosome
D	Polypeptide	RNA	Ribosome	RNA polymerase

- 14** The isolation of mRNA can be done by passing an RNA mixture obtained from homogenised tissue through a chromatography column. The procedure of the mRNA isolation is as follows:

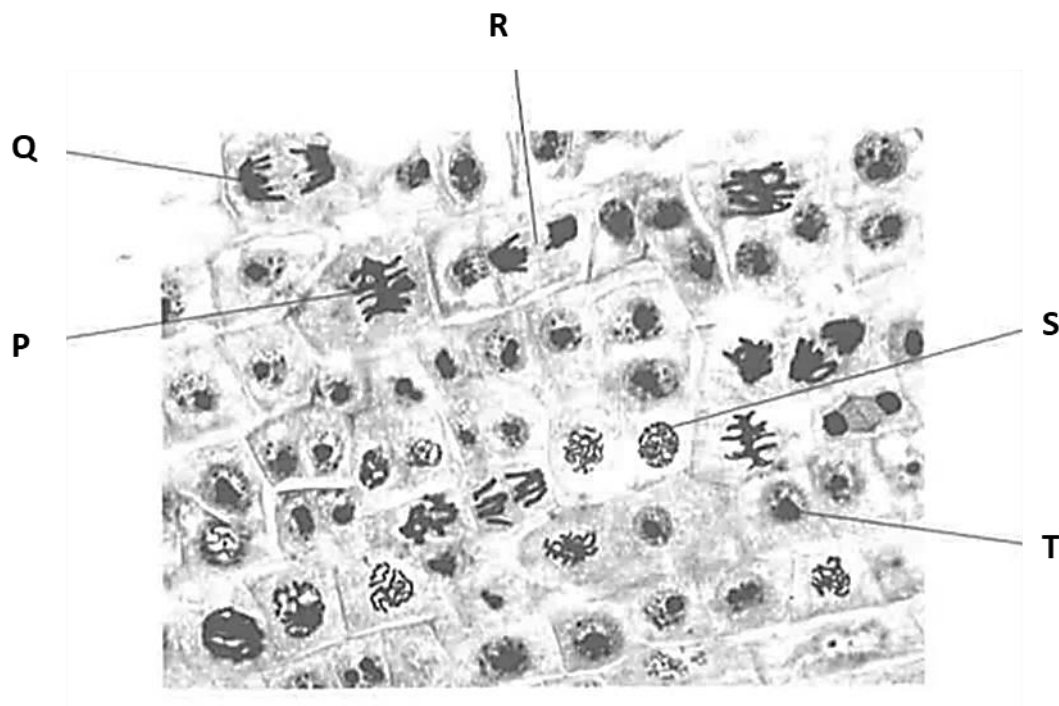


Step 1	Set up a chromatography column which contains short lengths of uracil nucleotides attached to a solid support medium.
Step 2	Add RNA mixture to the chromatography column. RNA that do not hybridise with the uracil nucleotides will pass through and leave the column.
Step 3	Add water to the chromatography column to remove and isolate the hybridised mRNA.

Which of the following statements is **not** true?

- A** rRNA and tRNA molecules could pass through and leave the column in Step 2.
- B** Complementary base pairing is responsible for the attachment of RNA to the solid support medium.
- C** The RNA that attached to the chromatography column lack poly-(A) tails.
- D** The isolated mRNA in Step 3 do not contain introns.

- 15 The photomicrograph shows a plant tissue sample prepared from a growing root tip.



Match the cells with the correct stages in mitosis.

	prophase	metaphase
A	T	Q
B	S	P
C	T	P
D	R	Q

- 16 Meiosis is a form of cell division, used in sexual reproduction.

What is the particular importance of meiosis to living organisms?

- A** It is a very rapid process.
- B** It results in genetically identical daughter cells.
- C** Each division produces a large number of cells.
- D** It reduces the number of chromosomes in each cell.

17 These statements aim to describe gene mutation.

1. It can occur in both somatic and sex cells.
2. It can cause sickle-cell anemia and Down syndrome in humans.
3. It can affect mRNA splicing if mutation occurs within the intron of the gene.
4. It can change a dominant allele into a recessive allele, but not a recessive allele to dominant allele.

Which statements are true?

- A** 1 and 3
B 1, 2 and 3
C 1, 3 and 4
D 1, 2, 3 and 4

18 Which options correctly show the type of mutations that can occur in each of these phases of the cell cycle and whether the mutation can be passed down in the gametes?

	Phase of cell cycle	Type of mutation that can occur	Mutation is passed down to gametes
1	mitosis	Polyploidy	✗
2	S phase	Gene mutation	✗
3	meiosis	Aneuploidy	✓
4	S phase	Chromosomal translocation	✓

- A** 1 and 3
B 1, 2 and 3
C 3 and 4
D 1, 2, 3 and 4

19 Which statement does **not** explain how cancer arises?

- A** Mutations resulting in the loss of contact inhibition.
- B** Increased expression of genes that promotes cell division.
- C** Loss of stem cells with age.
- D** Error during cell division that resulted in duplication of cell cycle promoting genes.

20 A group of scientists had recently used gene editing technique to delete a defective gene that is responsible for a fatal heart condition, in human embryonic stem cells.

What could be a possible ethical concern?

- A** The use of embryonic stem cells might result in the eventual death of an individual.
- B** The use of embryonic stem cells suggest the prevention of the embryonic stem cells from using the cells for reproduction.
- C** The successful use of gene editing techniques suggest the possibilities of creating designer babies.
- D** The gene editing technique is new and expensive, available only to affluent group of scientists.

21 Which observation would rule out a sex-linked trait in a pedigree?

- A** Male-to-male inheritance
- B** Male-to-female inheritance
- C** Males expressing the disease
- D** More females expressing the disease

- 22** A naturally occurring mutant tomato plant with yellow fruit was crossed with the red wild type.

The first generation plants were self-pollinated and were also backcrossed with both parents. The results are below.

- The first generation plants all produced red coloured fruits.
- The second generation plants produced fruits in a ratio of 3 red : 1 yellow.
- Plants from a backcross with the yellow mutant produced fruits in a ratio of 1 red : 1 yellow.
- The mutant yellow fruit had a higher rate of chlorophyll breakdown during fruit ripening.
- The red fruit from the first generation showed lower rates of chlorophyll breakdown as similarly observed in red wild type fruits.

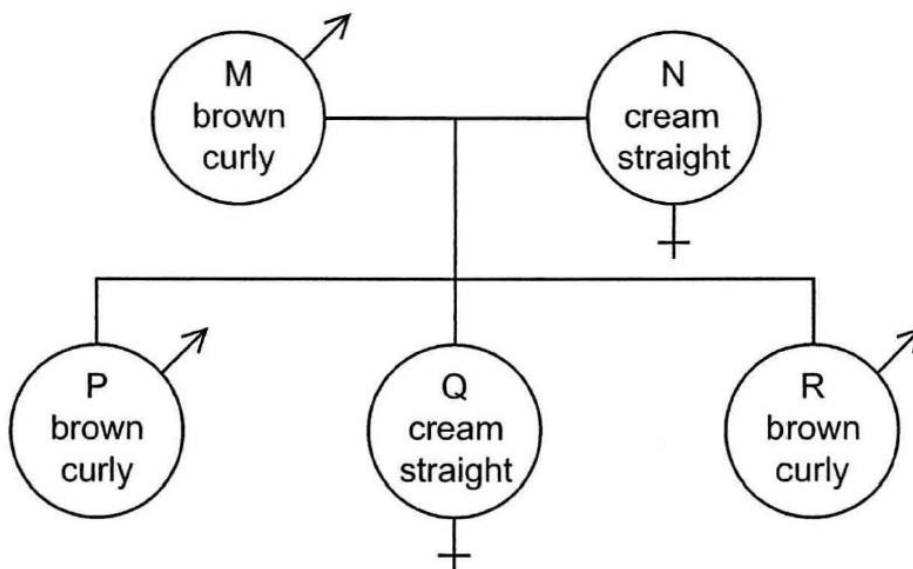
What can be concluded from the results above?

1. Yellow phenotype is coded for by a recessive allele.
2. The dominant red allele influences chlorophyll metabolism in fruits.
3. The effect of the red allele on chlorophyll metabolism is inhibited by the yellow allele.

- A** 1 only
B 1 and 2
C 1 and 3
D 2 and 3

- 23** Assume that in goats, a pair of alleles is responsible for the inheritance of hair colour and that another pair controls hair texture. These pairs are located on different autosomes. The allele for brown hair is dominant to allele for cream hair, and the allele for curly hair is dominant to the allele for straight hair.

The diagram shows a cross between two goats.



If R is mated with a female goat of the same genotype as M, what are the chances of the first offspring being a male with cream coloured, straight hair?

- A** 0
 - B** 1 in 4
 - C** 1 in 16
 - D** 1 in 32
- 24** Which of the following processes involve the use of NADH as a electron donor?
- A** Calvin cycle
 - B** Krebs cycle
 - C** Oxidative phosphorylation
 - D** Photophosphorylation

25 A student made the following claims on photosynthesis:

- Decrease in light intensity results in a decrease in rate of Citric acid cycle
- ATP from substrate level phosphorylation is required for light-independent reaction to take place
- Photosynthesis involves the conversion of light energy to chemical energy
- NADH is formed in photophosphorylation

How many of the student's deductions are correct?

- A** 1
B 2
C 3
D 4

26 Glucose, made from six radioactively-labelled carbon atoms, is fed to yeast cells in the presence of oxygen.

Which row shows the correct location and number of the identified radioactively-labelled product?

	Location	Radioactively-labelled product	Number of radioactive molecules per glucose
A	Mitochondrial matrix	Pyruvate	3
B	Cytosol	Acetyl-CoA	3
C	Mitochondrial matrix	Carbon dioxide	6
D	Mitochondrial intermembrane space	ATP	34

- 27** Yeast cells were obtained and the cell membrane was removed. The contents of the cell were then placed in a test tube. The contents were incubated with NADH without oxygen.

5 cm³ of oxygen was then added and some observations were recorded.

Which is a possible observation?

- A** Carbon dioxide can be detected before and after the addition of oxygen.
 - B** There is an initial rise in pH in the cytosol.
 - C** There is an initial drop in pH in the mitochondrial intermembrane space after the addition of oxygen.
 - D** Ethanol was detected before oxygen was added.
- 28** Deer mice are small, ground dwelling rodents. They normally have dark coats. Some paler coated mice have been observed in an area where new sand hills were formed between 8 000 and 15 000 years ago. The sand hills are sparsely covered with scrubby plants. Studies have shown that the change in coat colour was due to a mutation in a gene, causing a single amino acid deletion from the protein for the coat colour pigment. This mutation seems to be spreading through the population.

Which statement most fully explains how the evolution of this species at this site has occurred?

- A** Better camouflage increases the chance of survival by reducing predation.
- B** The occurrence of the mutation provided new variation on which natural selection can act.
- C** The paler colour gives better camouflage against the background of the new sand hills.
- D** Deer mice which live longer usually leave more offspring since they have more reproductive opportunities.

29 Which of the following is **not** the consequence of natural selection?

- A** The field mustard plant survived a summer drought in southern California because some individuals contained alleles that made them flower earlier. Plants with flowers wilt more easily than plants without flowers. Now, almost all the field mustard plants in California flower in spring.
- B** In areas with fewer predators of herbivorous insects, plants which produce higher concentrations of alkaloids (which are toxic to insects) dominated the landscape. Most of the herbivorous insects in these areas are found to be able to accumulate alkaloids in their bodies without affecting their metabolism.
- C** Endemic to New Zealand, the kakapo (a large flightless bird) had no natural predators before the humans arrived. They have evolved to have very few offspring throughout their entire lifespan. This phenomenon is also common for other island species which do not have natural predators in their respective habitats.
- D** Maple probably has the most variation in bark of any tree species. Japan experiences tornadoes which destroy large trees like the maple. Over the last few decades, it was observed that only the Japanese maple with dark-coloured bark remained.

30 Which is true about climate change?

- A** It is a result of human activities only.
- B** It is a result of temperature changes.
- C** It is a result of enhanced greenhouse effects.
- D** It is a result of ice cap melting and extreme weather patterns.

Y6 H1 P1 PRELIM

Answers

Qn	Ans	Qn	Ans	Qn	Ans
1	A	11	B	21	D
2	C	12	B	22	B
3	D	13	A	23	D
4	A	14	C	24	C
5	D	15	B	25	A
6	A	16	D	26	C
7	B	17	A	27	C
8	D	18	A	28	B
9	B	19	C	29	D
10	A	20	C	30	C